Zeyad Mohamed Salem

Full-Stack Developer · React, TypeScript, Supabase/Postgres · Alexandria, Egypt
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Available remotely for freelance and part-time work, up to 20 hours per week.

SUMMARY

Full-stack developer and Computer Science student at E-JUST. I build web apps end to end, from requirements and database design to security review, testing and production, with one app in daily use since September 2026. I enforce authorization in Postgres with row-level security and back my work with more than 2,900 automated tests in CI. I build bilingual Arabic/English interfaces, and I use AI coding tools such as Claude Code to deliver faster while owning the architecture, decisions and quality. Co-author of an IEEE NILES 2026 paper.

EDUCATION

B.Sc. Computer Science, Egypt-Japan University of Science and Technology (E-JUST) Expected December 2027

PUBLICATION

A. A. Alshames, I. Abdelmotteleb, M. Sameh, **Z. M. Salem**, S. Farrag, G. Morcos, S. A. Alashry, A. B. Zaky, "AraBIRD: A Modern Standard Arabic Adaptation of BIRD for Text-to-SQL over English Schemas," *2026 IEEE 8th Novel Intelligent and Leading Emerging Sciences Conference (NILES)*. Accepted.

PROJECTS

Samboza: Bilingual Household Finance App (PWA) · **github.com/Zeyadmosalem/Samboza** 2026 · In production
React 19 · TypeScript · Vite · Supabase (Postgres) · Cloudflare Workers · GitHub Actions · Vitest · pgTAP

- Built and shipped an installable Arabic/English web app with a full right-to-left layout and dark theme, used daily by a five-person household since September 2026.
- Ran the full development cycle: interviewed users for requirements, designed the data model, reviewed security, wrote test plans, and deployed.
- Enforced authorization in the database with Postgres row-level security across four user roles, proven by 532 pgTAP policy tests plus 578 front-end tests running in CI on every push.
- Implemented a double-entry ledger with an audit trail, offline writes through an idempotent outbox, Excel export, and CI/CD across 50 migrations that blocks deploys until the schema is current.

NC Spark: Role-Based Compliance Training Platform · **github.com/Zeyadmosalem/nc-spark** 2026
React 19 · JavaScript · Vite · Supabase (Postgres 17, Edge Functions, Realtime) · TanStack Query · Cloudflare Workers · Vitest

- Built a learning platform with four portals (trainee, trainer, supervisor, admin) and seven activity types, from quizzes and flashcards to scenarios and file submissions, all authored in the browser.
- Secured assessments by grading quizzes in Edge Functions against an answer key no browser role can read, and blocked privilege escalation with column grants, row-level security and triggers, verified by red-team test suites.
- Placed the site behind a Cloudflare Worker access gate that checks sessions, throttles sign-ins and sets security headers (CSP, HSTS); 1,828 tests at 91% coverage across 39 migrations and 14 Edge Functions.

Beyond Seen Mistakes: Research Pipeline (test & CI contributor) 2026
Python · automated testing · GitHub Actions

- Wrote the test suite and CI for a published research pipeline: data-loader, split-protocol, training edge-case and reproduction tests, and made reproduction checks numerical so they pass reliably across machines.

TRAINING

Mobile App Development Intern, DEPI (Digital Egypt Pioneers Initiative) · Mobile track, final project: a weather app 2024

iOS Development Intern, ITI (Information Technology Institute) · Native iOS development with Swift 2024

SKILLS

Languages: TypeScript, JavaScript, SQL, Python, C++, Dart, HTML/CSS
Front end: React, Vite, Tailwind CSS, TanStack Query, offline-first PWAs, right-to-left (Arabic) interfaces
Back end & data: Supabase (Postgres, row-level security, Edge Functions, Realtime), Cloudflare Workers, Node.js, database migrations
Testing & DevOps: Vitest, pgTAP, Testing Library, pytest, red-team security tests, GitHub Actions CI/CD, Git
Tools: AI-assisted development (Claude Code)
Spoken: Arabic (native), English (professional working proficiency)